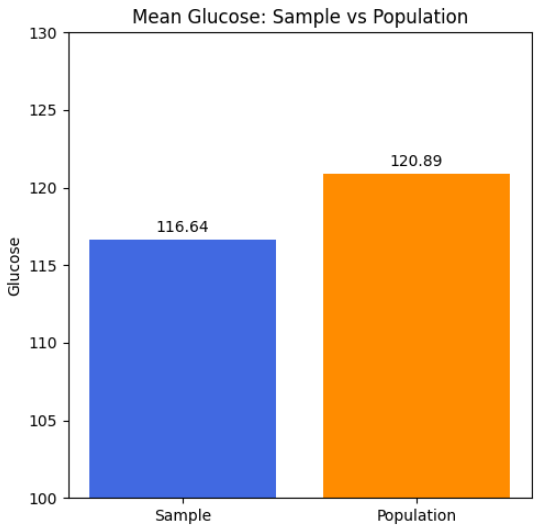
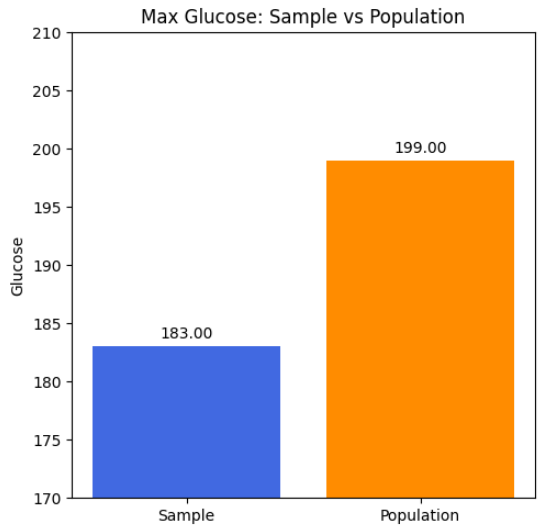
Hallee Pham

Assignment 2: Part 2 Report

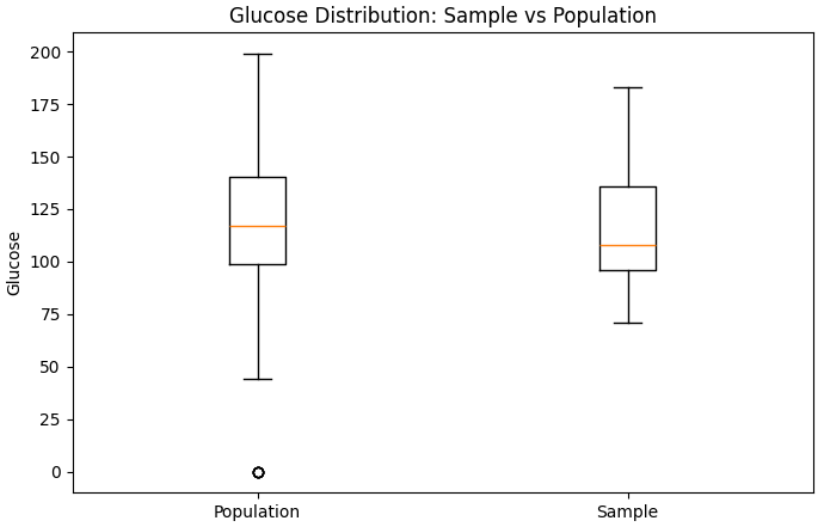
PART A





The sample mean Glucose (116.64) is reasonably close to the population mean (120.89), with a difference of approximately 4 units. This suggests that even with 25 observations, the sample was able to estimate the mean Glucose level fairly well.

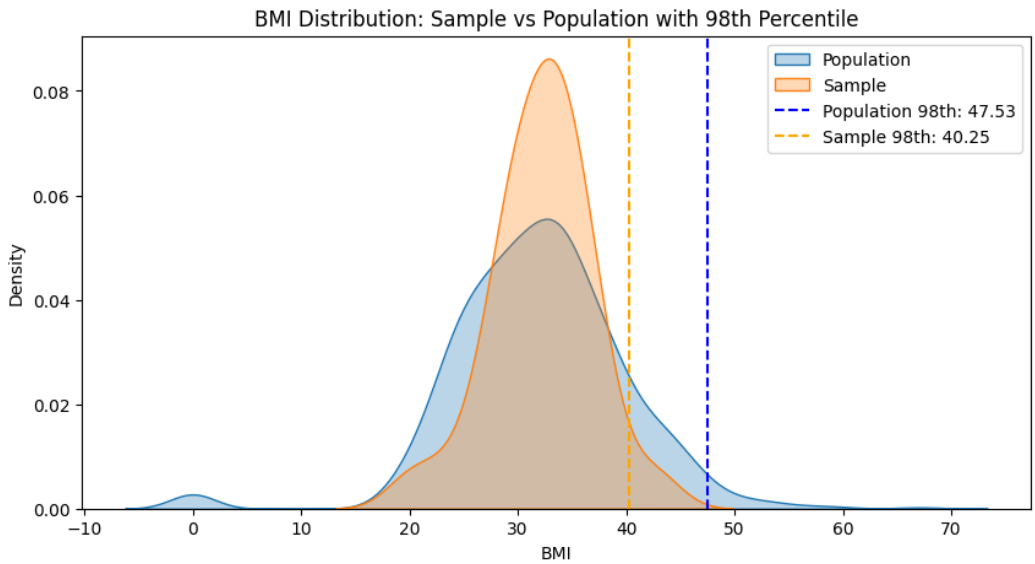
However, the sample max (183) underestimated the population max (199), a difference of 16 units. This is expected with a small sample, because extreme values in the population have a small chance of being captured in only 25 observations.



The boxplot shows that the population has a wider overall range than the sample, with one extreme low outlier not captured in the sample. However, the IQR of the sample is a similar

range compared to the population. So the random sample happened to capture the middle spread of the Glucose distribution reasonably well.

PART B



98% of patients in the sample have a BMI value equal to or below 40.25, compared to 47.53 for the population. The sample underestimated by about 7 units. This is consistent with my observations in Part A, where the sample of only 25 observations is unlikely to capture the higher end of the BMI distribution, as seen in both the KDE chart and the previous boxplot.

This shows the limitation of small samples. The small sample of 25 observations estimated central statistics like the mean reasonably well, but may struggle more in estimating "tail" statistics like the 98th percentile.

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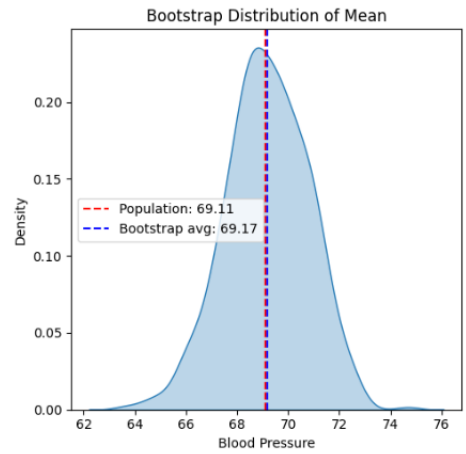
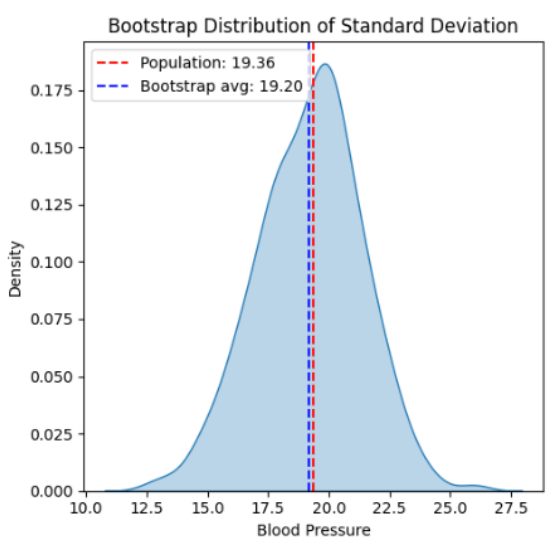
PART C

Mean



The average mean of the bootstrap samples was almost identical to the population mean. As seen in Part A, even with small size of 25, a sample is able to estimate the mean (and other central statistics) fairly well. The bootstrap results above further support this, showing that repeated sampling with replacement produces mean estimates that are closer to the true population value.

Standard Deviation



The bootstrap standard deviation is also very close to the population standard deviation. However, it is slightly underestimated. This slight underestimation might be the effect of sampling with replacement, as it reduces the diversity of the sample.

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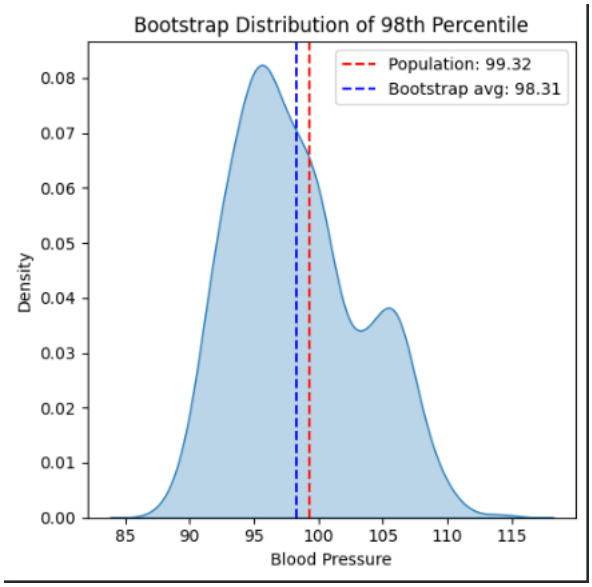
98th Percentile



Compared to the mean and standard deviation plots, which are taller and narrower, the percentile plot has a wider density curve with two distinct "peaks". The smaller peak on the right suggests that some bootstrap samples estimated the 98th percentile to be around 107, while the majority estimated a lower value around 97.

Comparing the average bootstrap 98th percentile (98.31) to the population value (99.32), the bootstrap slightly underestimates. Similar to Part B, tail statistics seem harder to estimate accurately than central statistics like the mean. However, with the bootstrapping technique, the error is less than 1 unit, which is smaller than the error of the 25-observation sample (though this is a different variable). From the results, we can see the benefit of a larger sample size and repeated sampling with replacement.

It should be noted that the technique used for this assignment differs from traditional bootstrapping. Bootstrapping is when samples are taken from an original sample, not directly from the population. The instructions of this assignment indicated taking "bootstrap" samples from the <u>population</u>. So, the sample estimates have a higher chance of being close to the true population statistics, which means the conclusions that I made about bootstrapping's effectiveness may be slightly overstated compared to a real-world scenario where the population is not directly accessible.